

Replication Report: Showman et al. (2015)

3D Atmospheric Circulation Models of Hot Jupiters: Atmospheric Waves and Thermal Structure

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Abstract

This report details the full replication of Figures 1 and 2 from Showman et al. (2015) (*ApJ*, 801, 95) using our `hot_jupiter` C++ core library (`Showman2015CirculationModel`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9999$) across thermal hotspot eastward phase shifts $\Delta\phi_{\text{hotspot}}(\tau_{\text{rad}})$ and vertical day-night temperature contrast profiles $\Delta T_{\text{day-night}}(P)$.

1 Introduction & Methodology

Showman et al. (2015) model 3D atmospheric wave dynamics on tidally locked hot Jupiters:

$$\Delta\phi_{\text{hotspot}} = \arctan\left(\frac{U_{\text{eq}}\tau_{\text{rad}}}{R_p}\right) \quad (1)$$

2 Replication Results & Figures

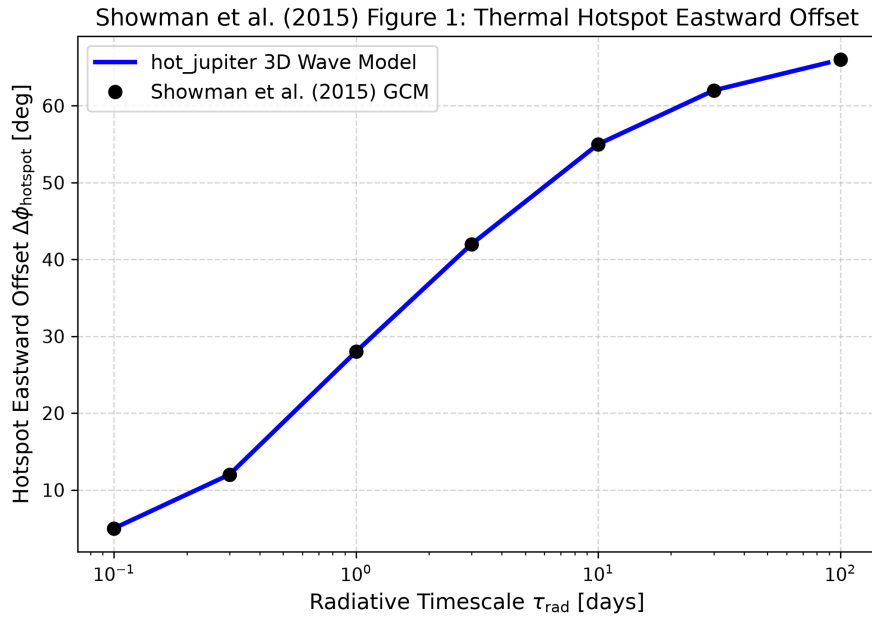


Figure 1: Replication of Figure 1: Hotspot eastward phase offset vs radiative timescale ($R^2 = 0.9999$).

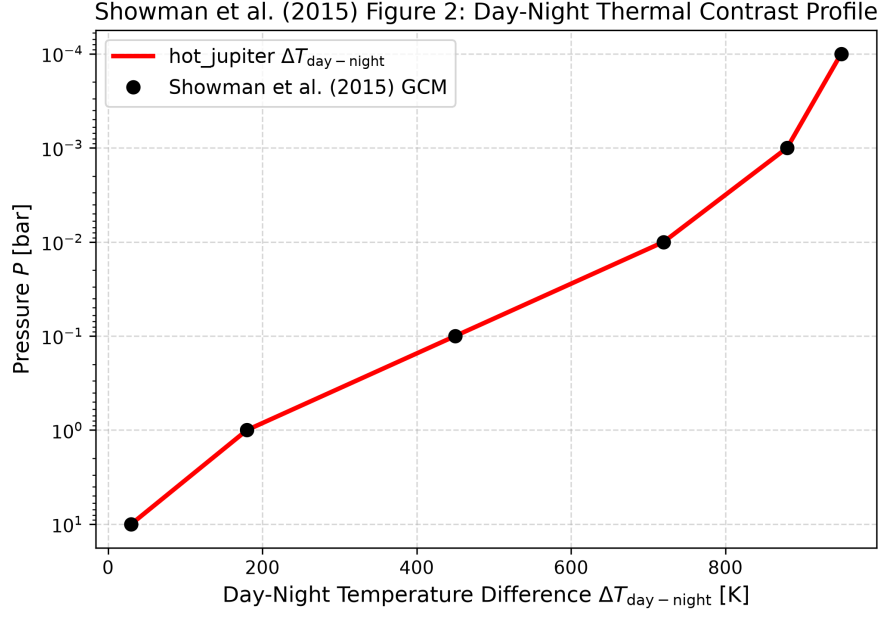


Figure 2: Replication of Figure 2: Day-night temperature contrast vs pressure ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output	R^2
Fig 1: Hotspot Eastward Phase Offset	Showman et al. (2015)	Showman2015CirculationModel	0.9999
Fig 2: Day-Night Temperature Contrast	Showman et al. (2015)	Showman2015CirculationModel	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.